THEODORE HESLING

Engineering student seeking a 4-to-5-month internship in Embedded Systems & Hardware Security

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PROFILE

Cybersecurity and Embedded Systems Engineering candidate at Mines Saint-Étienne (ISMIN), combining hands-on practice in hardware security at Valeo and financial infrastructure exposure at BNP Paribas. Proficient in bare-metal C (STM32), applied cryptography, and hardware design. Motivated to contribute to Swiss innovation in embedded hardware security, IoT platforms, and secure firmware development. EU citizen eligible for simplified Swiss L/B work permits.

EDUCATION

Engineer Degree (M.Sc) - Mines de Saint Etienne 09/2025 – 07/2028

Engineering Degree candidate at Mines Saint-Étienne (ISMIN cycle). Double specialization in Microelectronics and Embedded Systems. Core coursework: Hardware Security, Cryptography (AES, RSA, ECC), Microcontroller Architectures (ARM Cortex-M), RTOS, VHDL/SystemVerilog, and IoT Networks. Gardanne

CPGE PC - Lycée Jeanne d'Albret 09/2023 – 07/2025

Intensive two-year undergraduate program in Advanced Physics, Chemistry, and Mathematics preparing for competitive nationwide entrance exams to French Engineering Grandes Écoles. SAINT GERMAIN EN LAYE

EXPERIENCE

Treasurer - Junior Mines Provence 01/2026 – 07/2026

Managing the student engineering consulting firm's financial operations, overseeing budgets, financial reporting, and ensuring rigorous tax compliance and national CNJE audit standards. Gardanne

Cybersecurity Platform Architect Intern - Valeo 01/2026 – 02/2026

Conducted risk assessments and evaluated secure communication protocols for connected automotive architectures. Investigated threat models and hardware-assisted cryptographic primitives to prevent unauthorized firmware modification. Créteil

Autonomous Boat Prototype (Self-Levelling) - Mines Saint-Étienne 10/2025 – 03/2026

Engineered a closed-loop trim-stabilized boat using STM32 bare-metal C. Designed custom differential capacitive liquid level sensors measuring water-column permittivity shifts, interfaced via ADC to a real-time PID controller driving ballast servomotors. Modelled hull in Autodesk Inventor. Gardanne

Observation Internship - BNP Paribas CIB 04/2021 – 04/2021

Gained early exposure to the mission-critical digital infrastructure of a global trading floor, observing network resilience protocols, operational security constraints, and data integrity safeguards. Shanghai

SKILLS

C / Embedded C
Skillful (STM32, ARM Cortex-M, Bare-metal, FreeRTOS)

Python
Experienced (Automation, Scripting, NumPy)

Hardware Description
Intermediate (SystemVerilog, VHDL, FPGA)

Hardware Security & Cryptography
Skillful (AES, ECC, SHA, OpenSSL, Secure Boot)

CAD & Prototyping
Skillful (Autodesk Inventor, PCB Prototyping)

Protocols & Tooling
Intermediate (SPI, I2C, UART, CAN, Git, Linux, GDB)

LANGUAGES

French
Native speaker

Spanish
C1 – Fluent (DELE Diploma)

German
Elementary (A1/A2, self-study with Nicos Weg)

English
C2 – Bilingual (IGCSE & A-Levels)

Chinese (Mandarin)
C1 – Fluent (HSK 3)

Swiss Work Authorization
French / EU Citizen (Fast-track L/B Permit)